

A Plugin-Mediated Migration Assessment from Azure Databricks to OCI AI Data Platform

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¹Coursework lab / AIDP Migration Workbench evaluation

Abstract

This paper reports a fictitious-data migration assessment designed to evaluate whether the *AIDP Migration Workbench* plugin can support a repeatable Azure Databricks to Oracle Cloud Infrastructure (OCI) AI Data Platform (AIDP) workflow. The source workload contained two Azure Databricks objects, `customers_src` and `orders_src`, and a joined output, `orders_customers_joined`. The workflow used project-local Azure and OCI contexts, a manifest-driven AIDP contract, wrapper commands exposed through `codex-oci`, browser evidence, and functional validation in AIDP Workbench. The final result preserved row-count parity across all observed stages: Azure Databricks count = 1000, OCI Object Storage count = 1000, AIDP Workbench count = 1000, and absolute difference = 0. The experiment is accepted for the tested join-preserving scenario and identifies one plugin improvement: adding `aidp platforms delete` for complete wrapper-mediated teardown.

1 Introduction

Cloud data-platform migrations often fail not because a dataset cannot be copied, but because the process is poorly evidenced: source objects are not inventoried, migration assumptions remain implicit, validation is incomplete, or cleanup is left outside the operational workflow. This assessment evaluates *AIDP Migration Workbench*, a Codex plugin distributed as `oci-cli-workbench`, under a controlled Azure Databricks to OCI/AIDP scenario.

The report is written as an evidence-focused scientific article rather than a marketing demonstration. Its structure prioritizes a concise abstract, formal tables, framed figures, reproducibility artifacts, and a terminal bibliography tied to the migration tools used in the assessment.

The main research question is:

Can a joined Azure Databricks workload be transferred and validated in OCI/AIDP using the *AIDP Migration Workbench* workflow, preserving exact row-count parity across Azure Databricks, OCI Object Storage, and AIDP Workbench?

2 Plugin Under Test

The plugin under test is *AIDP Migration Workbench*, internally named `oci-cli-workbench`, version 0.1.12. Its public repository is:

jgangini/codex-plugin-oci-aidp-migration-workbench

Folder	Role
<code>.az/</code>	Project-local Azure CLI and Databricks context.
<code>.oci/</code>	Project-local OCI CLI config for live OCI/AIDP work.
<code>.source/</code>	Portable source evidence, exports, SQL, notebooks, and notes.
<code>aidp/</code>	AIDP manifests, reports, and runtime handoff artifacts.

Table 1: Local context folders used by the migration workflow.

The Azure and OCI contexts were kept separate to avoid mixing credentials or cloud-specific assumptions.

2.1 Installation and Use

The plugin can be installed from the public GitHub repository using the Codex plugin marketplace command with the repository slug. When a fixed version is required, use the same marketplace add command with the pinned version suffix v0.1.12.

For local testing before publishing, clone the plugin repository, enter the cloned plugin directory, and run the local marketplace add command from that directory. The installed marketplace entry can later be upgraded with the Codex plugin marketplace upgrade command for the AIDP Migration Workbench entry.

The experiment established the following practical use sequence:

1. Prepare `.az/` for Azure inventory and Databricks source context.
2. Keep OCI configuration in `.oci/`.
3. Place exported SQL, notebooks, CSV files, and notes in `.source/`.
4. Build or repair `aidp/config/migration_manifest.json`.
5. Run `codex-oci aidp source inspect` and `codex-oci aidp manifest inspect`.
6. Inspect platform and compute state with `aidp platforms get` and `aidp compute list/ensure`.
7. Create the target bucket and upload the source manifest plus the joined CSV through `aidp storage` commands.
8. Validate the row count in AIDP Workbench.
9. Clean user-managed compute and temporary storage through the wrapper.

3 Experimental Design

The assessment used fictitious data and excluded development time, login time, general cloud-service provisioning, and unrelated cleanup waiting time from benchmark timing. The measured scope was source export, plugin-mediated transfer, functional validation, and evidence capture.

Object	Rows	Role
customers_src	100	Customer dimension source
orders_src	1000	Order fact source
orders_customers_joined	1000	Joined migration output

Table 2: Source and joined objects used by the migration assessment.

3.1 Acceptance Criteria

The migration is accepted when Azure Databricks returns `COUNT(*) = 1000`, the exported Azure CSV contains 1000 rows, the OCI Object Storage download contains 1000 rows, the AIDP Workbench notebook reads the plugin-uploaded object and returns 1000 rows, and the absolute difference across the comparison is 0.

4 Plugin-Mediated Execution

The corrected final run did not rely only on console clicks. Browser evidence documented live cloud state and the AIDP notebook output, while the migration transfer and cleanup path were represented through `codex-oci` command summaries. Table 3 lists the wrapper-level operations captured in the final evidence package.

Command	Observed result
<code>codex-oci aidp source inspect</code>	Source ready; Databricks hints present.
<code>codex-oci aidp manifest inspect</code>	Manifest valid after repair.
<code>codex-oci aidp platforms get</code>	AIDP platform Active.
<code>codex-oci aidp compute list</code>	Workspace and compute visible.
<code>codex-oci aidp compute ensure --dry-run</code>	Validation compute reusable.
<code>codex-oci aidp storage buckets create</code>	Temporary bucket created.
<code>codex-oci aidp storage objects put --location-key source-manifest</code>	Source manifest uploaded.
<code>codex-oci aidp storage objects put --location-key compat-export</code>	Joined CSV uploaded.
<code>codex-oci aidp storage objects list</code>	Manifest and CSV visible.
<code>codex-oci aidp compute cleanup --apply</code>	Temporary compute deleted.
<code>codex-oci aidp storage buckets delete --empty</code>	Bucket emptied and removed.

Table 3: Plugin and wrapper command evidence for the corrected migration run.

5 Evidence

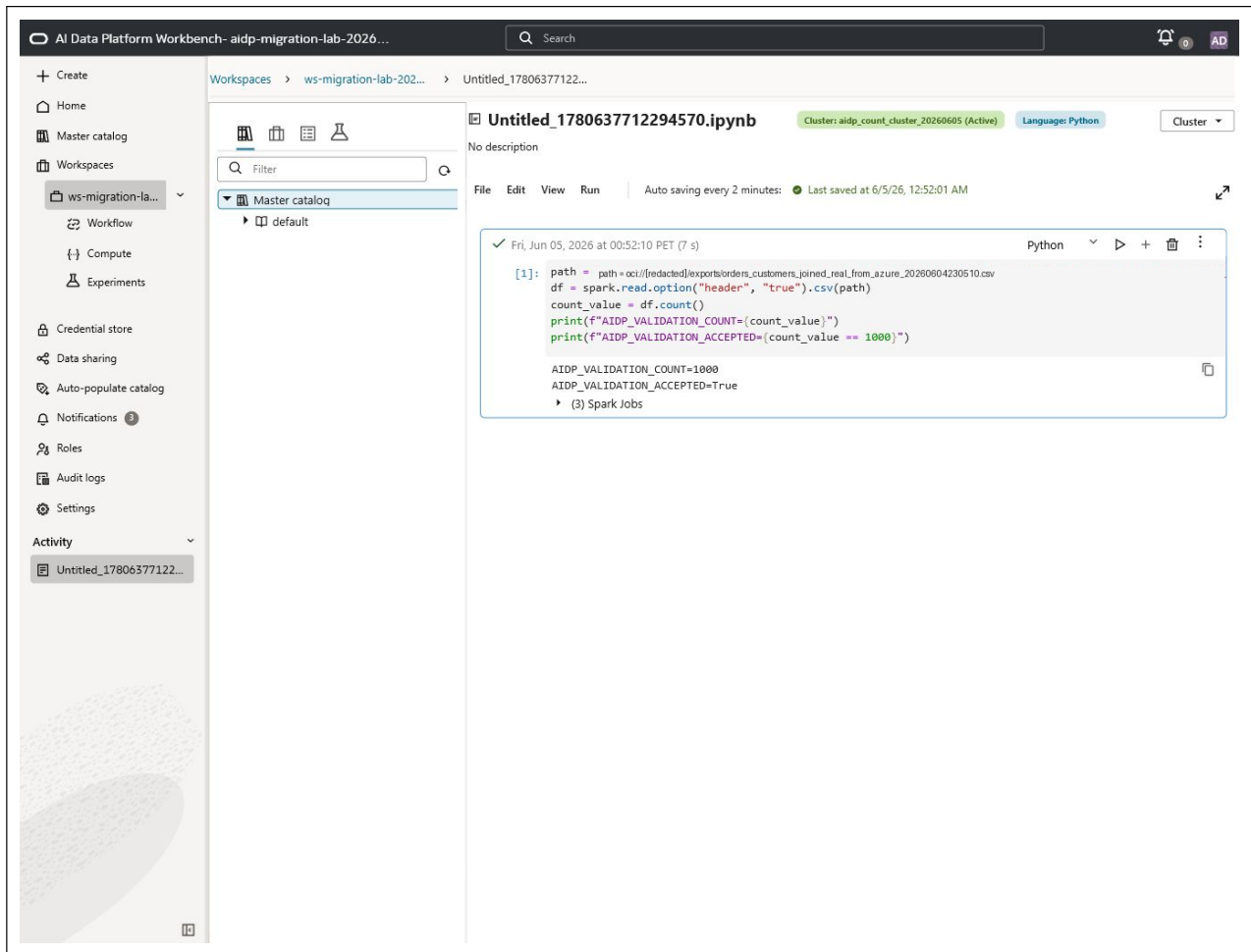


Figure 1: OCI AI Data Platform Workbench console capture showing notebook count validation output.

Codex Plugin Chat Evidence - Validation Chat

Structured Spanish test question and accepted plugin result

Usuario

Nueva validacion funcional posterior a la correccion completa.

Actua como usuario evaluador del plugin oci-cli-workbench / AIDP Migration Workbench en este proyecto.

Usa solo evidencia local del experimento en target/azure-oci-aidp-migration-lab. No inventes recursos cloud.

Debes responder:

1. Que comandos del plugin/wrapper prueban que la migracion no fue solo manual?
 2. Que objetos se migraron desde Azure Databricks?
 3. Como se valida que el join conserva el conteo en Azure, OCI Object Storage y AIDP Workbench?
 4. Que resultado final obtuvo la prueba?
 5. Que limpieza se realizo y que brecha queda en el plugin?
- Cita los archivos locales que prueban cada respuesta.

Codex

Validacion funcional final: aceptada.

Azure Databricks = 1000; OCI Object Storage = 1000; AIDP Workbench = 1000; diferencia = 0.

Wrapper/plugin: aidp source inspect; aidp manifest inspect; aidp platforms get; aidp compute list/ensure; aidp storage buckets create; aidp storage objects put/list; aidp compute cleanup; aidp storage buckets delete.

Objetos: customers_src (100), orders_src (1000), orders_customers_joined (1000).

Brecha: falta aidp platforms delete para teardown completo.

Figure 2: Codex validation chat evidence showing the structured test question and accepted result.

Codex Plugin Chat Evidence - Visible Tool Trace

Codex thread observations after AIDP validation

Codex visible messages

- Voy a hacer una nueva pasada de evidencias: busco senales explicitas de comandos del wrapper/plugin y de validacion en AIDP Workbench, ademas de Azure y OCI.
- La correccion completa dejo artefactos nuevos: plugin_execution_summary, source_manifest, capturas 47-51 y PDF final completo.
- La brecha tecnica local ya esta clara: el wrapper cubre inspeccion, manifest, computo, storage y limpieza de compute/bucket; la eliminacion de la plataforma AIDP uso OCI CLI base porque el wrapper no expone aidp platforms delete.
- Para la parte visual de AIDP abri la captura clave del notebook para confirmar que el texto local coincide con los JSON.

Tool trace

- mem_current_project executed in the Codex validation thread.
- mem_context executed for project continuity.
- semble semantic search executed against the current lab workspace.
- mem_save persisted the final plugin validation evidence summary.

Figure 3: Codex thread trace showing wrapper-command and AIDP Workbench validation evidence discovery.

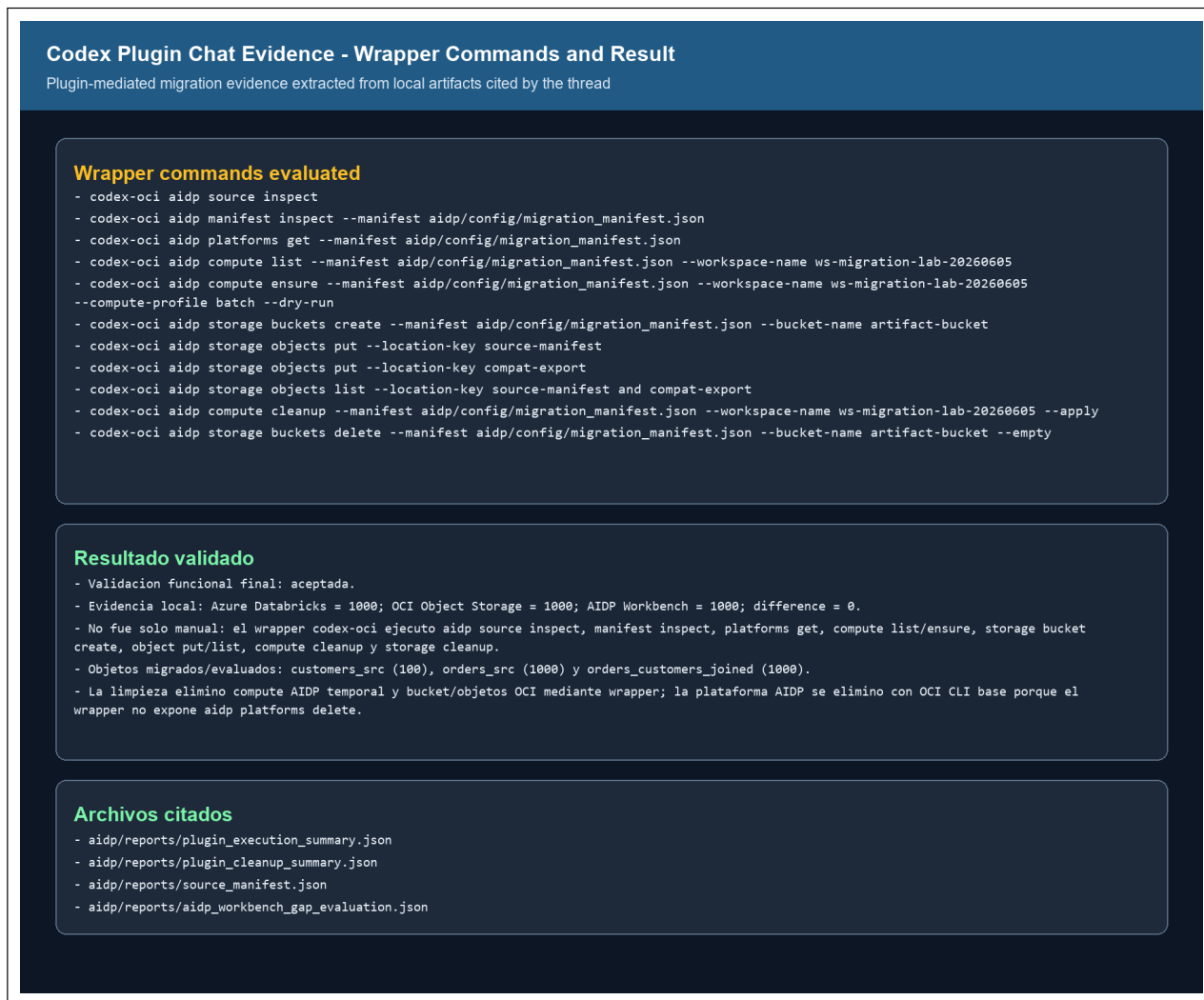
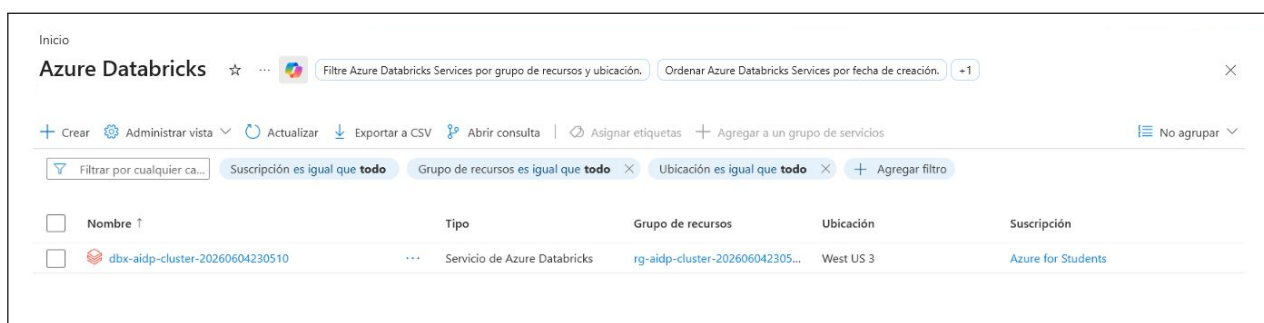


Figure 4: Codex validation result listing wrapper commands, accepted counts, cleanup, and the remaining plugin gap.



Azure Databricks Query History Evidence

Workspace	dbx-aidp-cluster-20260604230510
Warehouse	Serverless Starter Warehouse (RUNNING during validation)
Schema	codex_migration_sql_20260604230510
Objects	customers_src=100, orders_src=1000, orders_customers_joined=1000
Join object	orders_customers_joined created from orders_src INNER JOIN customers_src
Count query	<pre>SELECT COUNT(*) AS azure_join_count FROM codex_migration_sql_20260604230510.orders_customers_joined</pre>
Statement state	SUCCEEDED / Query History state: Finished
Statement ID	[redacted]
Observed Azure count	1000
OCI downloaded count	1000
Accepted difference	0
Redaction	User names and email addresses redacted; tokens never printed.

Figure 6: Databricks Query History evidence showing the joined-object count query in Finished state with observed count 1000.

Details

Objects

Management

Monitoring

Policies

Work requests

Tags

Objects

Q exports/

✕

Search

Upload objects

Actions ▾

☐

☐ Name/Prefix	Last modified	Size	Storage tier	
☐ .. (Go back)	—	—	—	...
☐ orders_customers_joined_real_from_azure_20260604230510.csv	Jun 5, 2026, 04:19 UTC	51.56 KiB	Standard	...

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Figure 7: OCI Object Storage evidence for the exported joined CSV.

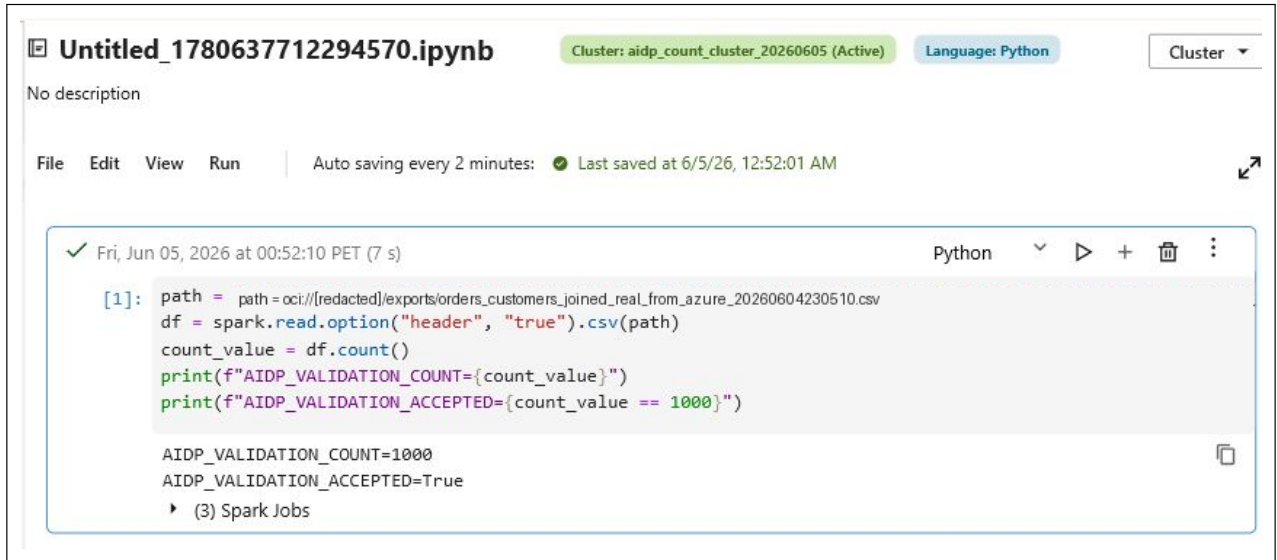


Figure 8: AIDP Workbench notebook output showing `AIDP_VALIDATION_COUNT=1000`.

6 Functional Validation

The final validation result is shown in Table 4. All row-count observations converge to the same value, proving exact parity for the tested joined output.

Validation item	Value
Azure Databricks joined count	1000
Exported Azure CSV rows	1000
OCI Object Storage downloaded rows	1000
AIDP Workbench count	1000
Absolute difference	0
End-to-end accepted	Yes

Table 4: Functional count validation across Azure Databricks, OCI Object Storage, and AIDP Workbench.

Note. Deployment time, login time, exploratory debugging, and unrelated cleanup waiting were excluded from benchmark timing. Only export, migration, validation, and evidence capture belong to the measured assessment scope.

7 Discussion

The most important correction in the experiment was separating service provisioning instability from migration validation. Earlier attempts encountered compute-capacity limits. The accepted run used a working Databricks SQL path and then validated the migrated object in OCI/AIDP. This distinction matters scientifically: provisioning time and exploratory debugging should not be counted as migration runtime, but they should be documented as operational risk.

The experiment also clarified the plugin’s current boundary. The wrapper supports source inspection, manifest inspection, platform and compute inspection, storage creation, object upload, object listing, compute cleanup, and storage cleanup. However, full platform teardown required OCI CLI base because the wrapper does not yet expose `aidp platforms delete`. That gap does not invalidate the migration result, but it is a concrete product improvement.

8 Cleanup

Resource	Final observed state
Temporary AIDP user compute	Deleted through the wrapper cleanup command.
Temporary OCI bucket and objects	Removed through the wrapper storage cleanup command.
Temporary AIDP platform	Deleted through OCI CLI base fallback
Temporary Azure/Databricks resources	Cleanup evidence recorded

Table 5: Cleanup outcomes for temporary resources created by the assessment.

9 Conclusion

The corrected migration assessment is accepted. The evidence shows that *AIDP Migration Workbench* and its `codex-oci` workflow were used to inspect source context, validate the manifest, create and upload OCI migration artifacts, list transferred objects, and clean temporary compute and storage. Azure Databricks, OCI Object Storage, and AIDP Workbench all returned a count of 1000 for the joined output, with absolute difference 0. The remaining recommendation is to add `aidp platforms delete` to the plugin so full teardown can be conducted without a base OCI CLI fallback.

Reproducibility Package

File	Purpose
<code>plugin_execution_summary.json</code>	Wrapper commands executed during transfer and validation.
<code>plugin_cleanup_summary.json</code> <code>source_manifest.json</code>	Temporary compute and storage cleanup evidence. Source objects, row counts, and validation query contract.
<code>migration_manifest.json</code>	AIDP migration target, paths, and runtime configuration.
<code>aidp_workbench_gap_evaluation.json</code>	Notes on AIDP notebook coverage and remaining plugin gap.
<code>final_completion_audit.json</code> <code>codex_thread_plugin_validation_redacted.json</code>	Final acceptance status and parity result. Spanish chat evidence for plugin-use validation.

Table 6: Named reproducibility files retained for the final assessment.

References

- [1] Oracle. Oracle Cloud Infrastructure CLI documentation. <https://docs.oracle.com/en-us/iaas/Content/API/Concepts/cliconcepts.htm>.
- [2] Microsoft. Azure CLI Reference. <https://learn.microsoft.com/en-us/cli/azure/?view=azure-cli-latest>.
- [3] Joel Gangini. AIDP Migration Workbench for Codex. <https://github.com/jgangini/codex-plugin-oci-aidp-migration-workbench>.